
Design of a Four-State Reconfigurable Metasurface Pixel

Industrial Internship Report

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Abstract

Reconfigurable reflectarray antennas (RRAs) offer a promising alternative to conventional phased arrays. By controlling a reflected wave rather than equipping every radiating element with a complete active radio-frequency chain, RRAs can substantially reduce hardware complexity, cost, and power consumption. Their performance nevertheless remains limited by the quantization and losses of the elementary reflecting cells, referred to here as *pixels*.

This report presents the design of a pixel providing four discrete reflection phase states. It first reviews the physical principles of electronically steered arrays and compares the main beamforming architectures. It then introduces Greenerwave's RRA architecture and establishes the relationship between a pixel's resonant response and its complex reflection coefficient. Finally, it describes the complete design workflow: selection of a microstrip topology, electromagnetic modelling in CST Studio Suite, optimization of a two-layer resonant structure, and integration of a single-pole four-throw (SP4T) switch. Full-wave simulations demonstrate the theoretical feasibility of four well-separated phase states while maintaining a usable reflection magnitude. The resulting design provides a credible basis for fabrication, experimental validation, and subsequent assessment at antenna level.

Keywords: reconfigurable reflectarray antenna, metasurface, beam steering, microstrip resonator, phase quantization, SP4T switch.

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1 Introduction

Wavefront control is central to many sensing and communication systems. A directive beam concentrates electromagnetic energy toward a receiver, thereby increasing the useful signal and reducing power radiated in unwanted directions. Conversely, the same principle can be used in reception to enhance spatial selectivity. Radar, satellite communications, cellular networks, and radio-frequency identification all rely on this ability to shape and steer an electromagnetic field.

Historically, wavefronts were controlled mainly through fixed passive devices, such as lenses, mirrors, and parabolic reflectors. Antenna arrays introduced a more flexible approach: interference between several radiating elements can produce a directional beam whose orientation is determined by their relative phases. Phased arrays extend this idea by controlling these phases electronically, enabling rapid steering without mechanical motion.

Modern active phased arrays provide excellent performance but require a large number of radio-frequency components. Their cost, power consumption, thermal management, and manufacturing complexity increase rapidly with the number of elements. RRAs seek a different compromise. A feed antenna illuminates a reconfigurable reflecting surface whose local phase response is electronically controlled. The metasurface then reshapes the incident wavefront into the required outgoing field.

The number of phase states available at each pixel is a decisive design parameter. Binary pixels are comparatively simple, but coarse phase quantization introduces aperture errors that reduce gain and increase side-lobe levels. Moving from two to four states should improve the approximation of the ideal continuous phase profile, provided that the additional states do not introduce prohibitive losses or excessive complexity.

The objective of this internship was therefore to investigate how a four-state pixel could be implemented for Greenerwave’s metasurface technology. The work addressed three questions:

- Which electromagnetic mechanisms govern the amplitude and phase of the pixel reflection coefficient?
- Which microstrip topology can provide four distinct, controllable phase responses within the target frequency band?
- Can this topology be combined with a realistic SP4T switch while retaining satisfactory RF performance?

2 Beam Steering with Phased Arrays

2.1 From an isotropic radiator to an antenna array

In a homogeneous medium, the field radiated by an ideal isotropic point source can be represented, in the frequency domain, by the scalar Green function

$$G(r, k) = \frac{e^{ikr}}{r}, \quad (1)$$

where r is the source–observation distance and $k = \omega/c$ is the wavenumber in free space.

For an array of N identical elements located at positions $\mathbf{r}_n$, linearity gives

$$\Psi(\mathbf{r}, k) = \sum_{n=1}^N A_n G(\mathbf{r} - \mathbf{r}_n, k) e^{i\phi_n}, \quad (2)$$

where A_n and ϕ_n are the excitation amplitude and phase of element n . In the far field, the geometrical path difference becomes a direction-dependent phase shift. The radiation pattern can therefore be written as the product of the element pattern and an *array factor*. For a uniform linear array with spacing d ,

$$AF(\theta) = \sum_{n=0}^{N-1} e^{in(kd \sin \theta + \Delta\phi)}. \quad (3)$$

The main beam points in the direction θ_0 for which the contributions add coherently:

$$kd \sin \theta_0 + \Delta\phi = 0. \quad (4)$$

Changing the progressive phase shift $\Delta\phi$ consequently steers the beam without moving the antenna.

Diagramme de Rayonnement des Phased Array 1D en fonction de N

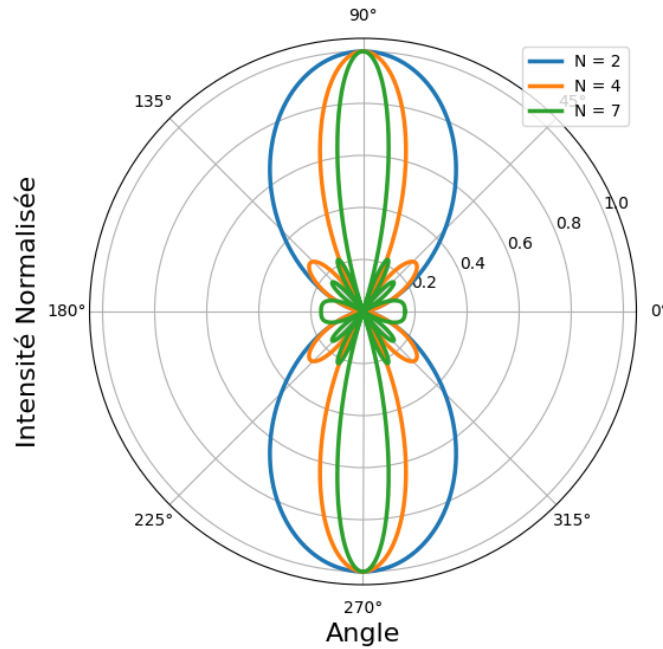


Figure 1: Simulated radiation patterns illustrating the increase in directivity as the number of elements in a uniform array is increased.

2.2 Use cases

Electronically steered antennas are commonly used in two operating modes. *Tracking* maintains the beam on a known moving terminal or satellite. *Scanning* sweeps the field of view to detect or identify targets. The former is central to mobile satellite and cellular links; the latter is typical of radar and inventory-tracking applications. Both require fast, repeatable steering and accurate control of the aperture phase.

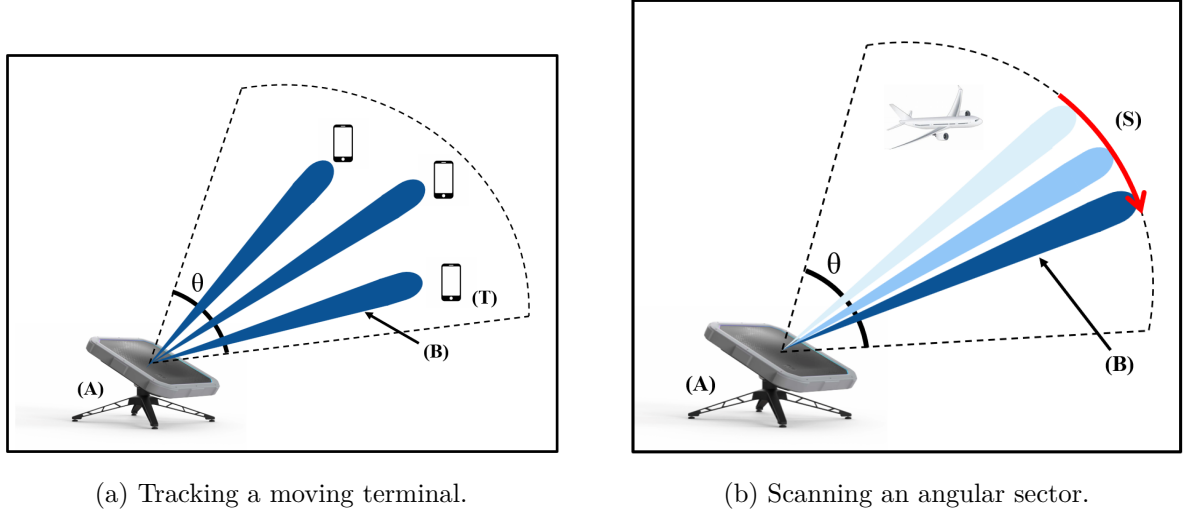


Figure 2: Two representative beam-steering operating modes.

2.3 Beamforming architectures

Three main architectures are used in conventional phased arrays.

Radio-frequency beamforming (RFBF) applies amplitude and phase control directly in the RF path. It is power-efficient and requires relatively few data converters, but generally supports a limited number of simultaneous beams.

Digital beamforming (DBF) assigns a complete transmit/receive chain and data converter to each element. It provides maximum flexibility, including multi-beam operation and frequency-dependent processing, at the cost of substantial power consumption, data throughput, and hardware complexity.

Hybrid beamforming (HBF) divides the array into subarrays and combines analogue phase control with a smaller number of digital chains. It offers a practical compromise for high-frequency communication systems.

The appropriate architecture depends on the required bandwidth, number of beams, steering speed, aperture size, and available power budget. RRAs belong to the broader family of RF-domain beamforming solutions, but replace a large fraction of the active phase-control hardware with a reconfigurable passive surface.

3 Greenerwave Reconfigurable Reflectarray Antenna

3.1 Antenna architecture

Greenerwave's antenna comprises three principal subsystems:

- a **feed antenna**, which illuminates the reflecting aperture;
- a **metasurface**, formed by a two-dimensional array of electronically reconfigurable reflecting pixels;
- a **partially reflecting cover**, or radome, which forms an open cavity with the metasurface and also protects the internal electronics.



Figure 3: Greenerwave RRA architecture: metasurface, feed, cavity, and partially reflecting cover.

The feed launches a wave carrying the information to be transmitted. Each pixel reflects the local incident field with a state-dependent complex coefficient

$$\Gamma_i(\omega) = |\Gamma_i(\omega)|e^{i\Phi_i(\omega)}. \quad (5)$$

By selecting the state of every pixel, the controller approximates the phase profile required to produce the desired far-field beam. At the partially reflecting cover, part of the field is radiated while the remainder returns to the metasurface. These multiple interactions enhance the metasurface's control over the emitted field.

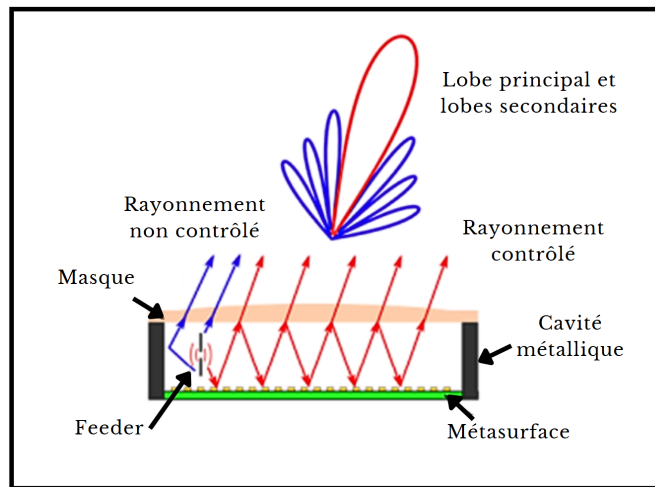


Figure 4: Operating principle of the reconfigurable cavity antenna.

3.2 Metasurface pixels

The metasurface is a planar printed circuit board populated with subwavelength reflecting cells. Each pixel combines a resonant microstrip structure with electronically controlled components. Its purpose is not to generate a signal, but to impose a prescribed phase shift on the reflected field while minimizing dissipation and undesired polarization conversion.

A microstrip patch illuminated by a linearly polarized plane wave behaves as an open electromagnetic resonator. The incident electric field induces a surface current. When the excitation frequency approaches a resonant frequency of the patch, successive reflections at its edges interfere constructively and a standing-wave pattern develops. In a simplified cavity model, the resonances scale approximately as

$$f_m \simeq \frac{mc}{2L_{\text{eff}}\sqrt{\varepsilon_{\text{eff}}}}, \quad m \in \mathbb{N}, \quad (6)$$

where L_{eff} and ε_{eff} account for fringing fields and the effective dielectric environment.

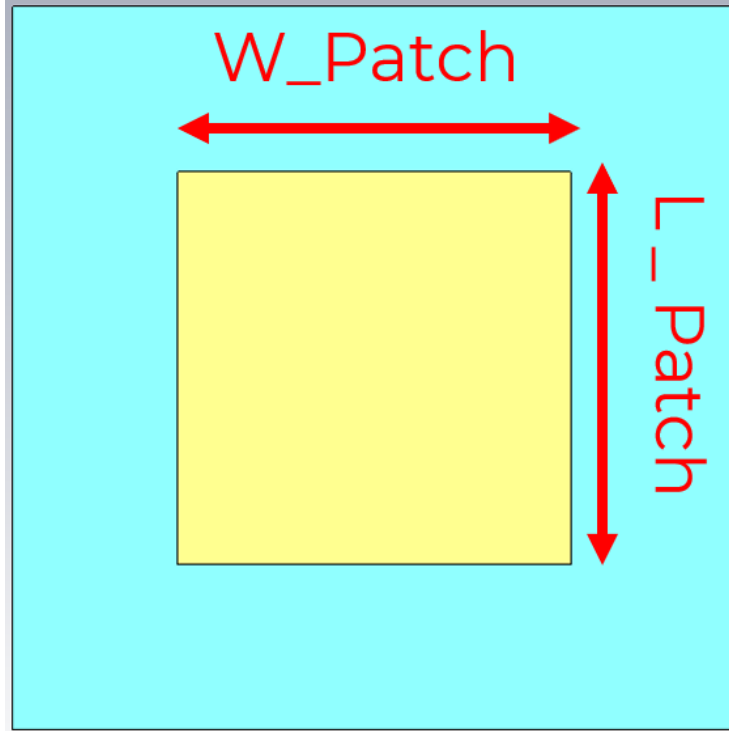


Figure 5: Representative microstrip patch used as the resonant element of a reflecting pixel.

3.3 Scattering description of the pixel

The scattering matrix relates the complex amplitudes of incoming and outgoing waves. For a two-port system,

$$\begin{pmatrix} B_1 \\ B_2 \end{pmatrix} = \begin{pmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{pmatrix} \begin{pmatrix} A_1 \\ A_2 \end{pmatrix}. \quad (7)$$

For the illuminated pixel, one channel represents the measured co-polarized plane wave, whereas additional effective channels account for radiation, ohmic, dielectric, and cross-polarization losses. The measured S_{11} therefore contains both the phase response required for wavefront control and the loss mechanisms that reduce antenna efficiency.

Near an isolated resonance, the reflection coefficient undergoes a rapid phase transition. Changing the geometry or electrical boundary conditions shifts the resonance and consequently changes the phase observed at a fixed operating frequency. Pixel reconfigurability is obtained by deliberately perturbing this spectrum. Two broad strategies are possible:

- modify the effective resonator geometry, for example by connecting or disconnecting conductive sections;
- couple the principal patch to a secondary resonant structure whose electrical length can be switched.

The second approach was selected because it separates the main reflecting element from the switching network and provides additional degrees of freedom for optimization.

4 Design of a Four-State Pixel

4.1 Performance criteria

A useful four-state pixel must be assessed against several simultaneous criteria. At the operating frequency, the four reflection phases should be well distributed over 360° , ideally with approximately 90° separation. The reflection magnitude should remain high and reasonably uniform across states. The response must also be robust over the required bandwidth and angular range, with limited cross-polarization. Finally, the geometry must be manufacturable and compatible with the selected switch, biasing network, substrate stack-up, and assembly tolerances.

These requirements are coupled. A narrow high- Q resonance produces a large phase excursion but is sensitive to tolerances and often provides limited bandwidth. Stronger coupling can broaden the response but may increase loss or make the four states difficult to balance. Optimization must therefore target the full complex reflection coefficient rather than phase separation alone.

4.2 Initial resonator selection

Several microstrip resonators were considered, including slots, split-ring resonators, and rectangular patches. The rectangular patch was retained as the main building block because its dominant mode is well understood, its resonance can be tuned primarily through its length, and it is straightforward to fabricate on a standard PCB stack-up.

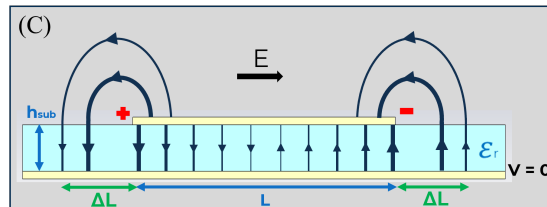


Figure 6: Simulated field or current distribution of the selected patch near its fundamental resonance.

The initial geometry was optimized in CST to position the resonance near the target band and to obtain sufficient phase sensitivity. Parameter sweeps were used to identify the influence

of patch length, width, substrate thickness, and inter-element spacing. This stage established a stable baseline before the switching network was introduced.

4.3 From two to four states

The first reconfigurable model used two alternative electrical lengths. It verified that a switched perturbing branch could move the resonance and produce two distinct reflection phases. The concept was then extended to four branches. Each branch corresponds to a different transmission-line length and therefore to a different impedance presented to the coupled resonator.

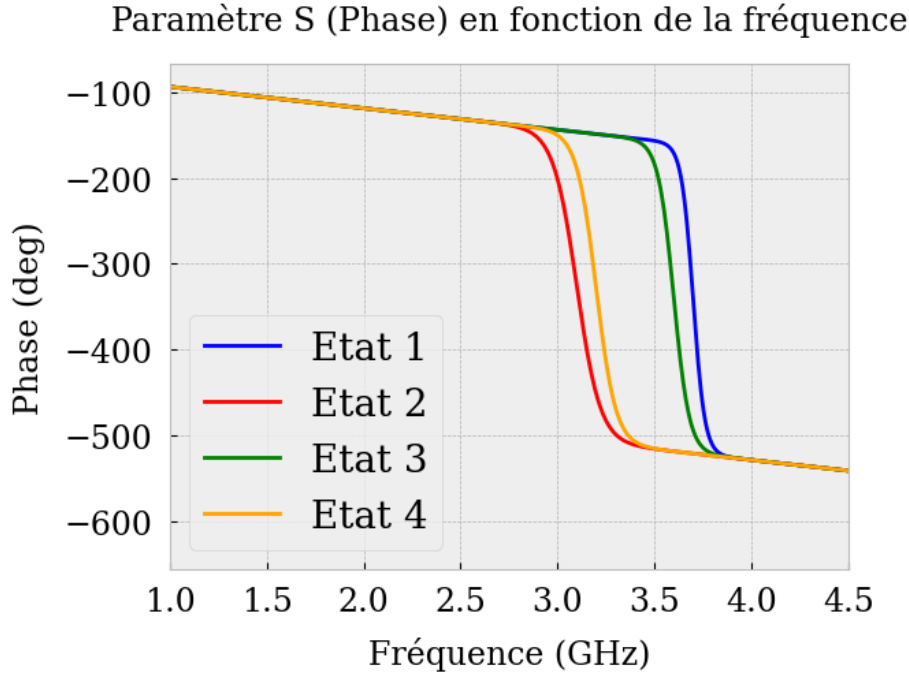


Figure 7: Simulated reflection phase of the four-state pixel configuration.

Ideal switching elements were used first to isolate the electromagnetic design problem. This model confirmed that four states could be obtained, but also showed that geometric transmission lines interact through parasitic coupling. The final line lengths therefore cannot be inferred independently from their ideal circuit values; they must be re-optimized in the complete three-dimensional model.

4.4 SP4T switch integration

A single-pole four-throw switch provides a compact means of selecting one of four branches. The component was introduced through its measured or manufacturer-provided state-dependent S-parameters. A circuit-level model was first used to verify port mapping and to translate each switch state into an equivalent transmission-line termination.

The switch is not ideal. Insertion loss reduces the reflected magnitude, while off-state isolation and parasitic reactances perturb the phase response. Interconnects, pads, vias, and reference planes also affect the effective electrical lengths. These contributions were progressively included to avoid optimizing a topology that could not survive physical implementation.

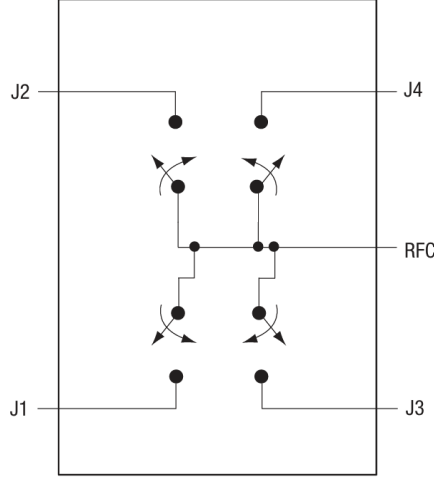


Figure 8: SP4T component used to select one of four electrical branches.

4.5 Two-layer coupled structure

The final concept uses two coupled resonant layers. The lower layer contains the four-state switching network, whereas the upper patch provides the principal interaction with the incident field. Coupling between the layers transfers the state-dependent perturbation to the reflection response.

This arrangement provides two important advantages. First, it reduces the direct loading of the radiating surface by the switch and its bias network. Second, the upper patch introduces an additional resonance that can be used to shape the phase slope and improve the trade-off between phase coverage, loss, and bandwidth.

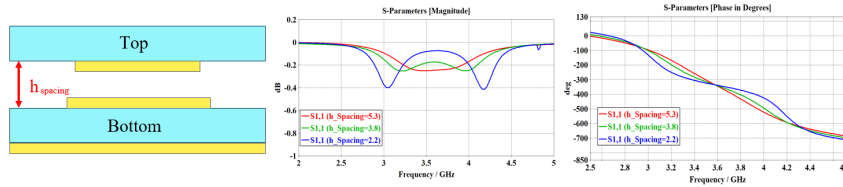


Figure 9: Coupling between the upper reflecting patch and the lower reconfigurable resonator.

4.6 Optimization workflow

The design was refined in three successive stages:

1. **Circuit-level optimization.** Ideal or schematic transmission lines were adjusted to obtain four approximately equidistant phases while constraining reflection loss.
2. **Independent 3D branch matching.** Each physical branch was optimized separately to reproduce its circuit-level response without coupling from the remaining branches.
3. **Full 3D co-optimization.** All branches, the SP4T model, and both resonant layers were simulated together. Line lengths and coupling parameters were then varied iteratively to compensate for mutual coupling and parasitic effects.

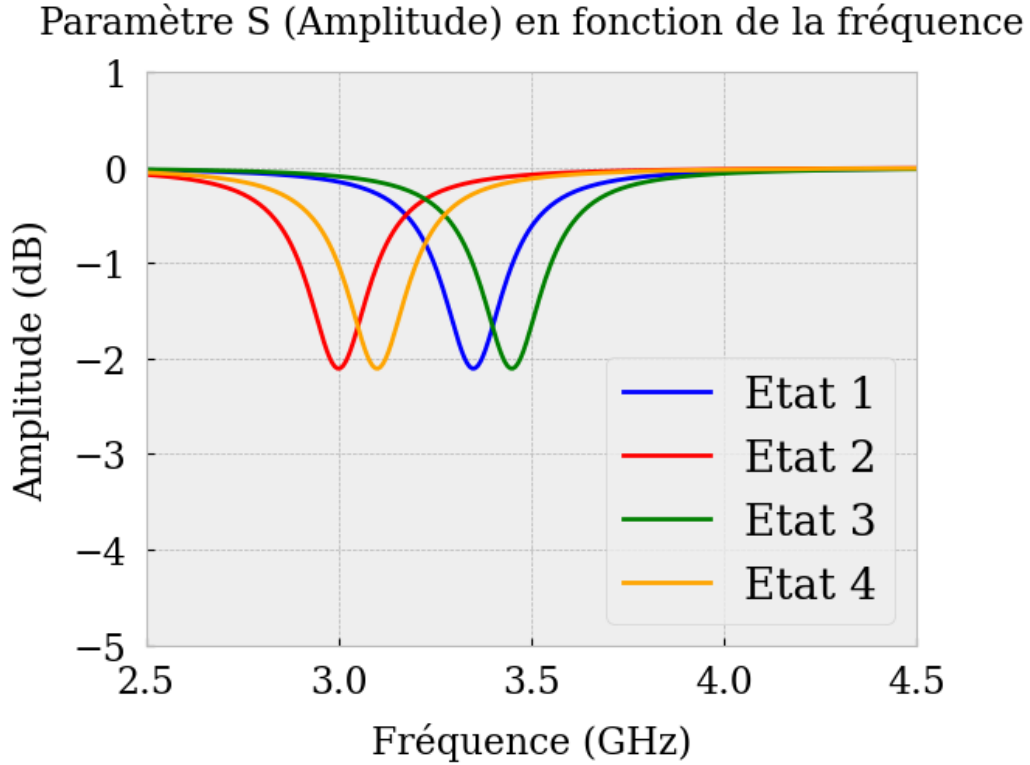


Figure 10: Representative optimized response of the complete two-layer four-state model.

The optimized simulations establish the feasibility of the concept: four distinguishable phase states are obtained around the design frequency with a reflection magnitude compatible with further experimental investigation. This result should nevertheless be interpreted as a design milestone rather than a complete antenna-level validation. PCB tolerances, connector and fixture effects, component dispersion, biasing, and calibration uncertainty must still be quantified experimentally.

4.7 Fabrication and measurement strategy

The prototype was prepared for fabrication using the material stack-up and assembly constraints available for the project. The measurement setup is designed to illuminate the pixel with a controlled incident field and recover the complex co-polarized reflection coefficient for each switch state. Calibration and reference measurements are required to remove the response of the antennas, cables, free-space path, and fixture.

5 Discussion and Next Steps

The present work demonstrates, in simulation, that a four-state pixel can be implemented using a coupled two-layer microstrip structure and an SP4T switch. The next step is experimental characterization of a fabricated unit cell. Measured phase separation, reflection loss, bandwidth, polarization purity, and sensitivity to incidence angle should be compared systematically with the full-wave model.

If unit-cell performance is satisfactory, the design should be integrated into a prototype metasurface. A controlled comparison between two-state and four-state operation on the same

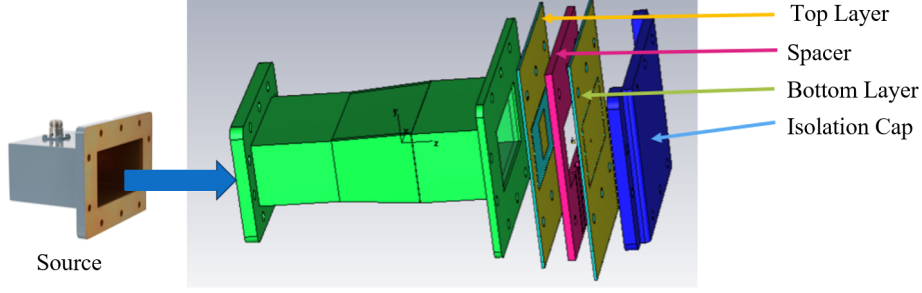


Figure 11: Proposed free-space measurement arrangement for characterizing the fabricated pixel.

aperture would isolate the value of finer phase quantization. Relevant antenna-level metrics include realized gain, aperture efficiency, beam-pointing error, side-lobe level, scan loss, and angular coverage.

The key engineering question is whether the reduction in quantization error outweighs the additional dissipation and complexity of the four-state implementation. The answer cannot be inferred from phase coverage alone. It requires an end-to-end comparison that includes switch loss, control electronics, manufacturing yield, calibration requirements, and the multi-reflection behaviour of the cavity.

Any discrepancy between simulation and measurement should be used to update the electromagnetic and circuit models. Particular attention should be paid to dielectric properties, copper roughness, component reference planes, soldering parasitics, and layer-alignment tolerances. This validated model could then support a second design iteration with lower dissipation, broader fractional bandwidth, and improved robustness. The same methodology could also be transferred to frequency bands relevant to satellite communications.

6 Conclusion

This internship investigated a central limitation of reconfigurable reflectarray antennas: the finite number of reflection phase states available at each metasurface pixel. After establishing the physical basis of phased array beam steering and positioning RRAs among existing beamforming architectures, the work focused on the resonant mechanisms that determine a pixel's complex reflection coefficient.

A complete design strategy for a four-state pixel was developed. It combined a coupled two-layer microstrip structure, four selectable electrical branches, and a realistic SP4T switch model. Progressive optimization—from ideal circuit elements to a full three-dimensional electromagnetic representation—was essential to account for coupling and parasitic effects. The final simulations support the theoretical feasibility of four distinct phase states and justify fabrication and measurement of a prototype.

Beyond the technical result, the project provided practical experience in electromagnetic modelling, simulation-driven design, and constrained optimization. It also illustrated the distinctive nature of industrial R&D: progress depends on balancing physical understanding, manufacturability, performance, and development time.

Working within a deep-tech company during a period of rapid growth offered a valuable perspective on the transition from exploratory research to a repeatable industrial product. The autonomy granted throughout the project strengthened my ability to formulate open-ended

problems, build appropriate models, and turn uncertain results into concrete design decisions. It also confirmed my interest in pursuing research at the interface between rigorous physical modelling and industrially relevant technology.

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